

$$\tilde{x}_{db} = \frac{x_{db} - \min_b x_{db}}{\max_b x_{db} - \min_b x_{db}}, \quad (5)$$

$$q_b(x_d) = \text{clip}(\lfloor \tilde{x}_{db} Q \rfloor, 0, Q - 1).$$

$$n_d^{V1}[b] = q_b(x_d), \quad |V_{V1}| = B. \quad (6)$$

$$n_d^{V3}[(b, q)] = \mathbb{1}[q_b(x_d) = q], \quad |V_{V3}| = BQ. \quad (7)$$

$$a_d = \arg \min_{a \geq 0, \mathbf{1}^\top a = 1} \|x_d - E a\|_2^2, \quad (8)$$

$$n_d^{V8}[m] = \min\left(\left\lfloor Q \frac{a_{dm}}{\max_{m'} a_{dm'}} \right\rfloor, Q - 1\right),$$

where $E = [e_1, \dots, e_M]$ are the NFINDR endmembers (simplex vertices), x_d is the unnormalised spectrum and $|V_{V8}| = M$. The abundance non-negativity (ANC, $a \geq 0$) and sum-to-one (ASC, $\mathbf{1}^\top a = 1$) constraints make a_d a point on the simplex spanned by the endmembers (fully-constrained least squares, FCLS; implemented as δ -augmented NNLS followed by renormalisation), so the coefficients are proper abundance fractions over the convex hull. The dominant endmember of every pixel emits $Q - 1$ tokens, so the document length grows with Q .

$$g_b(x_d) = \arg \max_{k \in \{0, \dots, Q-1\}} \pi_k \mathcal{N}(x_{db} | \mu_k, \sigma_k^2), \quad (9)$$

$$n_d^{V12}[(b, g)] = \mathbb{1}[g_b(x_d) = g], \quad |V_{V12}| = BQ,$$

where $(\pi_k, \mu_k, \sigma_k^2)_{k < Q}$ is a one-dimensional Gaussian mixture fitted once per scene to the unnormalised band intensities of the sampled pixels (a random subset of 50 000 values), with the components indexed in increasing order of μ_k . Each band emits one token, as in V3, but the bins are shared by all pixels and follow the intensity distribution instead of splitting each spectrum's own range into equal intervals.

$$w_b = \left\lfloor \frac{\text{MI}(x_b; y)}{\max_{b'} \text{MI}(x_{b'}; y)} Q_{\max} \right\rfloor, \quad (10)$$

$$n_d^{V20}[(b, q)] = w_b \mathbb{1}[q_b(x_d) = q].$$

The token alphabet is V3's, $|V_{V20}| = BQ$ (≈ 1600 at $B = 200, Q = 8$; mean ≈ 1376 across the six labelled scenes since B varies). Bands with $w_b = 0$ emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still (≈ 770 mean).

$$F-1 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2 P_c R_c}{P_c + R_c}. \quad (11)$$

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i, w_j) + \epsilon}{p(w_i)p(w_j)}}{-\log(p(w_i, w_j) + \epsilon)}, \quad (12)$$

$$c_v = \frac{1}{N} \sum_i \cos(\mathbf{v}_i, \sum_j \mathbf{v}_j),$$

with $\mathbf{v}_i = (\text{NPMI}(w_i, w_\ell))_{\ell=1}^N$ over the top- N topic words.

$$\hat{z}_d = \arg \max_k \theta_{dk}, \quad (13)$$

$$F-7 = \frac{I(\hat{Z}; Y)}{H(Y)} = U(Y | \hat{Z}) = 1 - \frac{H(Y | \hat{Z})}{H(Y)}.$$

$$F-14 = \frac{2}{K(K-1)} \sum_{k < k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \quad T_k = \text{top-}N(\phi_k). \quad (14)$$

$$F-18 = \frac{1}{\binom{S}{2}} \sum_{s < s'} \frac{1}{K} \max_{\sigma \in \mathfrak{S}_K} \sum_k \cos(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}), \quad (15)$$

over S reseeded refits, with $\mathbf{u}_k^{(s)}$ the top- N indicator vector of topic k in run s .

$$F-22(d) = \min_{\delta \in \mathbb{Z}^{|V|}} \|\delta\|_1 \text{ s.t.} \\ \arg \max_k \theta_k(n_d + \delta) \neq \arg \max_k \theta_k(n_d), \quad (16)$$

$$n_d + \delta \geq 0,$$

reported as the per-scene median, capped at a sentinel MAX_STEPS = 50 when no flip is found.

$$\text{aligned}(d) = \mathbb{1} \left[|\text{top}_N(n_d) \cap \text{top}_N(\phi_{z_d})| \geq \tau \right. \\ \left. \vee \text{top}_3(n_d) \cap \text{top}_N(\phi_{z_d}) \neq \emptyset \right], \quad (17)$$

$$\text{decisive}(d) = \mathbb{1} \left[\text{aligned}(d) = 1 \vee |\text{supp}(n_d)| \geq 3 \right],$$

$$F-15 = \sum_d \text{aligned}(d) / \sum_d \text{decisive}(d).$$

$$p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}}, \quad (18)$$

$$N_{\text{eff}}(d) = \exp(H(p_d)) = \exp\left(-\sum_v p_{dv} \log p_{dv}\right).$$

N_{eff} counts the tokens that effectively carry the mass of p_d (or of a topic's ϕ_k): it is low when the mass is concentrated on few tokens and at most $|V|$.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0, 1], \quad (19)$$

averaged over backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$; $\mathcal{A} \rightarrow 1$ when a recipe sits near the per-backbone top everywhere.

I. INTRODUCTION

The companion paper P3 [1] ranks the full V1–V20 wordification sweep (nineteen recipes; V16 is scaffolded only) on classification, coherence and label-coupling axes, and the backbone-factorial paper P4 [2] repeats the headline axes across four topic-model backbones. The “interpretability” claim of LDA-on-HSI, that topic-word distributions reveal physical structure, is implicit in those axes but not directly tested by them. *Throughout this paper we fix the backbone to LDA* (1); the cross-backbone behaviour of these interpretability axes is out of scope and left to P4. Three post-hoc interpretability tools have been proposed in the broader topic-modelling and explainable-ML literature that go beyond “read the top- N words and trust them”: SHAP attribution [3], counterfactual explanations [4], and LLM-as-judge alignment

[5]. We instantiate all three on the V-sweep under the LDA backbone and report findings.

The three axes correspond to three canonical metrics in the shared equation set: F-13 to the SHAP attribution applied to the closed-form LDA posterior, F-22 to the minimum- L_1 counterfactual (16), and F-15 to the top- N overlap rule (17). Our central methodological contribution is to show that F-15, an identity-overlap rule between a document’s top tokens and its topic’s top tokens, is governed by construction effects of the rule (vocabulary size, tokens per document and the sort order of tied counts) more than by agreement between documents and topics; we measure the topic-side dispersion, the effective vocabulary $N_{\text{eff}} = \exp(H)$ of (18), alongside it. The wordification recipes themselves are defined in P3; we recall only the four we lean on heavily: V1 (6), whose vocabulary has B word types, and V3 (7), V12 (9) and the label-aware V20 (10), whose shared vocabulary size ($B \cdot Q$) yet divergent N_{eff} make them the natural test of the cardinality-versus-dispersion question, a test that the rule’s document side leaves open.

A. The three axes

F-13 SHAP attribution. For each pixel document d with argmax topic z^* , attribute θ_{d,z^*} back to the recipe-specific tokens via SHAP’s KernelExplainer against the closed-form LDA posterior $p(z | x, \phi) \propto \exp(\sum_v x_v \log \phi_{z,v} + \log \alpha)$, the small-document specialisation of the LDA generative model (1). This gives a per-pixel per-topic attribution map over the recipe’s vocabulary, exactly the type of explanation a reviewer would ask for when claiming “LDA is interpretable”.

F-22 counterfactual L1. For each document d with argmax topic z^* , search via coordinate descent for the minimum- L_1 perturbation that flips the argmax from z^* to some $z' \neq z^*$, the discrete counterfactual of (16). Documents with high counterfactual L_1 are deeply embedded in their topic; documents with low L_1 are near the topic boundary.

F-15 LLM-as-judge alignment. For each sampled document, an LLM oracle is given the document’s top-10 tokens and the argmax topic’s top-10 tokens and asked whether the document semantically belongs to the topic; formally, the rule (17) with $N = 10$ and threshold $\tau = 3$ (§III). The F-15 values reported here are computed by a deterministic implementation of this rule that stands in for the live oracle (`build_v_sweep_f15_self_judge.py`); an implementation that queries an LLM through the Anthropic Messages API is provided in `build_v_sweep_f15_llm_alignment.py`.

II. RELATED WORK

Topic models on hyperspectral imagery. Probabilistic topic models were first applied to spectral data by Wahabzada et al. [6] (plant phenotyping) and to subpixel mixtures by Zou and Zare’s partial-membership LDA [7]. The CAOS_LDA_HSI series casts a spectral unit (here a pixel) as a “document” through a *wordification* recipe (5), starting from the band-frequency recipes V1–V3 introduced in [8], and runs online variational LDA [9], [10]. None of this prior work asks

whether the resulting topics are *post-hoc interpretable* beyond inspecting top- N words; that gap is the subject of this paper.

Topic-model evaluation. The dominant automated proxy for topic quality is coherence, NPMI / c_v [11]–[13] (12), together with stability under reseeding [14] (15). These reward internally consistent word lists but, as we show, are themselves sensitive to the vocabulary regime. We treat F-2 and F-7 (13) as the targets that our interpretability axes should *corroborate* rather than duplicate.

Post-hoc interpretability and explanation. SHAP [3] gives additive, locally faithful feature attributions with Shapley-value guarantees; we apply its KernelExplainer to the closed-form LDA posterior (F-13). Counterfactual explanations [4] characterise a decision by the minimal input change that flips it; F-22 (16) is the topic-assignment counterfactual. The use of large language models as automated topic judges follows Stambach et al. [5]; F-15 (17) is our LDA-on-HSI instantiation. Our contribution relative to this literature is the observation that an identity-overlap judge rule inherits construction effects of the wordification (vocabulary size, tokens per document, tied counts) that can dominate its score.

III. METHOD

Backbone, corpus and fits. Every result in this paper uses the LDA backbone (1) fitted with scikit-learn’s online variational solver [10], [15] (`max_iter` = 60, α = 0.45, β = 0.2, seed 42), the canonical fit of P3. The corpus is the six labelled scenes: Indian Pines [16], Salinas and Salinas-A [17], Pavia University [18], Kennedy Space Center [19] and Botswana [20]. Documents are pixel spectra; tokens are produced by the wordification recipes of P3 under the uniform quantiser (5) at $Q = 8$. We analyse the nineteen LDA-fitted recipes V1–V15, V17–V20 (V16, a foundation-model recipe, is scaffolded only).

F-13. We sample 50 documents per (V , scene) cell and use 25 background documents per SHAP KernelExplainer call against the closed-form posterior, reporting the top-8 tokens per topic by mean absolute SHAP value. The closed-form posterior is preferred over the amortised `transform()` because the latter needs internal state that cannot be restored from ϕ alone.

F-22. We compute the discrete minimum- L_1 counterfactual of (16) by coordinate descent over the integer token-count lattice, capped at a sentinel `MAX_STEPS` = 50; cells in which no sampled document flips persist at 50 rather than being dropped, so that “unflippable” bases are not silently treated as missing. We report the per-scene median over 20 sampled documents, then the mean across the six scenes.

F-15 and the dispersion metric. F-15 applies rule (17) to 20 sampled documents per (V , scene) cell, with $N = 10$ and $\tau = 3$. Here $\text{top}_k(n_d)$ lists the document’s k highest-count tokens among those with non-zero count; where equal counts straddle the cut, the list is not unique, $\text{supp}(n_d)$ is the set of tokens with non-zero count, and $\text{top}_N(\phi_{z_d})$ is the top- N list of the document’s argmax topic. A document is *aligned* if its top-10 tokens share at least three elements with the topic’s top-10, or if any of its top-3 tokens is among

the topic’s top-10 (this covers the sparse-event regime of V7 and V9, whose documents hold few distinct tokens). It is *misaligned* if it has at least three non-zero tokens and is not aligned, and *ambiguous* otherwise, that is, when it has fewer than three non-zero tokens and none of them is among the topic’s top-10. Ambiguous documents are excluded from the denominator: F-15 is the number of aligned documents divided by the number of aligned and misaligned ones. Exclusion affects V9 (36 of its 120 sampled documents) and V17 (26 of 120) and no other recipe. Because ties make the two top- N lists ambiguous, we evaluate each judged document under uniformly random tie-breaking, independently in the document and in its argmax topic, and report the exact expectation $\sum_d P(\text{YES}_d) / \sum_d (P(\text{YES}_d) + P(\text{NO}_d))$, computed by hypergeometric counting in rational arithmetic (`research_core/f15_alignment.py` of the code repository). The value equals the single-sort rule wherever no tie crosses a cut, and it does not change if the tokens or the topics are renumbered. Alongside F-15 we compute, for each topic-word distribution ϕ_k , the token-mass dispersion of (18),

$$N_{\text{eff}}(\phi_k) = \exp(H(\phi_k)) = \exp\left(-\sum_v \phi_{kv} \log \phi_{kv}\right),$$

the effective number of tokens carrying a topic’s mass. A low N_{eff} means the topic concentrates on few tokens, so its top- N is more likely to overlap a document’s top- N ; a high N_{eff} spreads mass thinly and depresses the overlap even when the topic is coherent. We report the per-recipe mean of $N_{\text{eff}}(\phi_k)$ over topics and scenes, paired with F-15 and $|\mathcal{V}|$ (Fig. 1). To show what the tie handling is worth, we also re-apply the single-sort rule to the same fits with ties broken by the archived order, by band index and at random (five seeds).

IV. F-13 SHAP RESULTS

Under the F-13 protocol of §III, the headline result is that SHAP attributions are *recipe-grounded*: a V1 topic’s top SHAP token always names a wavelength band; a V7 topic’s top SHAP token names an absorption feature “abs_c3_d4_a1” (centre, depth, area bucket); a V12 topic’s top SHAP token names a (band, Gaussian-component) pair. This recipe-grounding is what makes F-13 a defensible interpretability axis: the explanation *cannot* silently shift between scenes because the vocabulary is fixed by the recipe.

A sample from Indian Pines, recipe V7, topic 0:

```
abs_c3_d4_a1    mean|SHAP| = 0.0261
abs_c2_d0_a2    mean|SHAP| = 0.0201
abs_c3_d4_a2    mean|SHAP| = 0.0144
```

A geospectroscopist reading these can immediately ask: what is the wavelength range of centroid bucket 3? Why are depth bins 4 and 0 elevated? The answer is independent of which scene we apply V7 to.

V. F-22 COUNTERFACTUAL RESULTS

Table I reports, for each recipe, the median over 20 sampled documents of the minimum- L_1 perturbation that flips the argmax topic, averaged over the six scenes (§III).

TABLE I
F-22 MEDIAN COUNTERFACTUAL L_1 TO FLIP ARGMAX TOPIC, MEAN ACROSS 6 LABELLED SCENES, FULL 19-RECIPE SWEEP WITH THE SENTINEL PATCH (CELLS WHERE EVERY SAMPLED DOCUMENT FAILS TO FLIP WITHIN $\text{MAX_STEPS} = 50$ PERSIST AT 50 RATHER THAN BEING DROPPED). HIGHER MEANS MORE ROBUST TOPIC BOUNDARY.

Recipe	median L_1
V20	26.33
V12	24.50
V3	23.50
V2	7.83
V14	7.67
V1	6.08
V7	5.17
V15	4.00
V18	3.67
V8	3.58
V11	3.33
V13	2.67
V17	2.58
V4	2.50
V5	2.50
V19	2.08
V6	1.92
V10	1.17
V9	1.00

V20, V12 (24.5) and V3 (23.5) form an ultra-robust band at $Q = 8$: in a large fraction of the sampled documents no assignment flips within 50 steps (the sentinel), so the per-cell median is dominated by never-flipped scenes and the strict $V20 > V12$ ordering is an artefact of that sentinel patching; where flip-sampling is adequate (6–19 flipped documents per scene) V12 is the most robust. What the axis does support is that $V20 / V12 / V3$ are far harder to flip than the fragile recipes ($V9 = 1.0$). F-22 counts unit token changes, so it grows with the document length: on Indian Pines a V20 document holds 1187 tokens and a V3 or V12 document 200, against a single token for V9.

VI. F-15 LLM-JUDGE RESULTS: CONSTRUCTION EFFECTS OF THE RULE

The LLM-as-judge rule (17), with $N = 10$ and $\tau = 3$, marks a document *aligned* with its argmax topic if its top-10 tokens share ≥ 3 elements with the topic’s top-10 or if any of its top-3 tokens is among the topic’s top-10; F-15 is computed over the aligned and misaligned documents, excluding ambiguous ones (§III). Per-recipe mean F-15 across the 6 labelled scenes:

The ranking is *anti-correlated* with F-2 coherence on the large-vocabulary recipes V3 and V12, which P3 ranks at the top of F-2 and F-7. Three construction effects of the rule account for most of the table; Table III measures them by re-applying the rule to the archived fits (`scripts/f15_tie_sensitivity.py` of the manuscript repository, https://github.com/fsantibanezleal/CAOS_LDA_HSI_Paper).

(i) *Small vocabularies*. V2 (8 word types) and V8 (6–12) score 1.000 because a topic’s top-10 covers all or nearly all of the vocabulary, so any document token overlaps it.

TABLE II

F-15 LLM-JUDGE ALIGNMENT FRACTION (DETERMINISTIC STAND-IN JUDGE APPLYING THE RULE OF §III). NUMERIC VOCABULARY SIZES ARE MEANS OVER THE SIX SCENES, EXCEPT FOR V8, WHOSE VOCABULARY IS ITS SET OF 6–12 ENDMEMBERS PER SCENE. VALUES ARE THE EXPECTATION OF THE RULE UNDER UNIFORMLY RANDOM TIE-BREAKING (§III); THE 1.000 OF V2, V8 AND V9 IS FIXED BY THE RULE ITSELF (TABLE III).

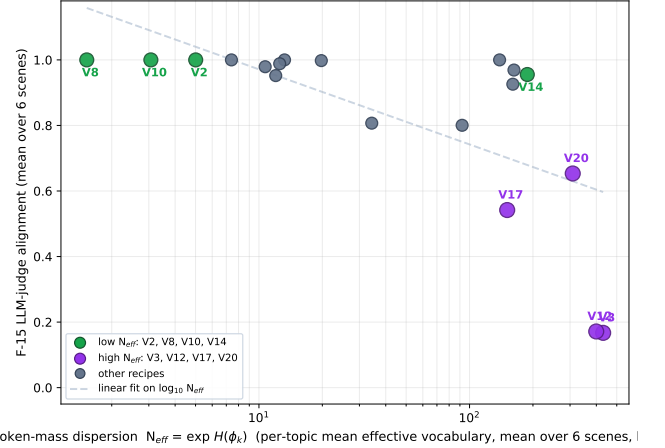
Recipe	F-15	vocabulary $ \mathcal{V} $
V2	1.000	8
V6	1.000	$\approx B$
V8	1.000	6–12
V9	1.000	$R \cdot Q$
V10	1.000	24
V19	1.000	24
V18	0.998	128
V15	0.988	44
V11	0.979	32
V5	0.969	B
V14	0.955	1024
V13	0.952	128
V4	0.926	B
V7	0.807	$\leq C_b Q^2$
V1	0.800	B
V20	0.653	BQ
V17	0.542	512
V12	0.171	BQ
V3	0.167	BQ

(ii) *Short documents.* A V9 document holds one token, and the rule can judge such a document aligned or ambiguous but never misaligned (a misaligned outcome needs at least three non-zero tokens), so V9’s 1.000 (84 aligned, 0 misaligned, 36 ambiguous) is fixed by the rule. The recipes with three or four tokens per document (V10, V11, V13, V19) score 0.95 to 1.00, since their whole document enters its own top-10.

(iii) *Tied counts.* V3 and V12 documents hold each of their 200 tokens once, so no sort order of the counts carries information about the document: every one of their judged documents is tie-dependent (119 and 120 of 120). A single sort therefore reports one arbitrary draw, between 0.033 and 0.180 for V3 and between 0.067 and 0.173 for V12, with the two recipes changing places between orders. Under the expectation of the rule they are 0.167 and 0.171, which is what Table II reports.

The one contrast that survives every tie order is V20 against V3 and V12 (0.653 against 0.167 and 0.171 under the expectation, 0.47 to 0.67 against at most 0.18 across single orders), at the same nominal vocabulary (mean $|\mathcal{V}| \approx 1376$; V3 and V20 share the identical (band, bin) alphabet). The effective vocabulary $N_{\text{eff}} = \exp(H)$ of (18) separates the three topic sets, $N_{\text{eff}}^{V3} = 431 > N_{\text{eff}}^{V12} = 400 > N_{\text{eff}}^{V20} = 309$, so V20’s topics concentrate their mass on fewer tokens. But V20’s per-band copies also act on the document side of the rule: a V20 document carries graded counts (0 to 8 copies per band), so its top-10 is made of its highest-copy bands, the same bands whose repeated tokens dominate V20’s topics, whereas V3 and V12 documents carry no such ranking. The document side of the rule therefore differs across the three recipes, and the V20 gap cannot be attributed to topic dispersion

F-15 alignment against token-mass dispersion
Spearman $\rho(\text{F-15}, N_{\text{eff}}) = -0.77$ (vs $\rho(\text{F-15}, |\mathcal{V}|) = -0.68$; see companion figure)



$N_{\text{eff}} = \exp(H(\phi_k))$ with ϕ_k the LDA topic-word distribution (mean over topics, then 6 scenes), from data/local/ly_sweep/lda_fit/s<scene>_<recipe>_uniform_QB(phi).npy; F-15 from data/derived/ly_sweep/f15_llm_alignment/. V2, V8 and V9 sit at F-15 = 1.0 by construction of the rule; V3 and V12 depend on the tie order of their equal-count documents (0.03 to 0.18).

Fig. 1. F-15 LLM-judge alignment versus per-topic mean token-mass dispersion $N_{\text{eff}} = \exp(H(\phi_k))$ (18), LDA backbone, six labelled scenes, $Q = 8$, nineteen recipes (Table II); the Spearman coefficients in the title are computed over the same nineteen, with average ranks for tied values. Point colours are categorical: green for the low- N_{eff} recipes (V2, V8, V10, V14), purple for the high- N_{eff} recipes (V3, V12, V17, V20), grey for the others. At the same alphabet size (mean nominal $|\mathcal{V}| \approx 1376$), V3 / V12 / V20 (purple, lower right) fall in dispersion ($431 > 400 > 309$) while F-15 is 0.17, 0.17 and 0.65; the V3 and V12 values are expectations over tie orders, and V2, V8 and V9 sit at 1.0 by construction (Table III), so the plotted association mixes dispersion with the rule’s construction effects.

alone. Across the nineteen recipes F-15 correlates negatively with N_{eff} (Spearman $\rho = -0.77$, average ranks for the tied values) and with $|\mathcal{V}|$ ($\rho = -0.68$); both correlations carry the construction effects above, since the small-vocabulary and short-document recipes have both a low N_{eff} and a forced or near-forced F-15.

The tie dependence is settled by the expectation above; what remains is that the rule still rewards small vocabularies and short documents. A defensible F-15 would also compare the observed overlap with the overlap expected for matched random token lists, and count a recipe only where the rule can return every outcome; a semantic embedding distance between token identifiers is a further option. We flag these as future work and recommend reading Table II as a description of how the rule behaves on each recipe, not as a ranking of topic interpretability.

VII. INTERPRETABILITY UNDER THE V13–V20 EXTENSION

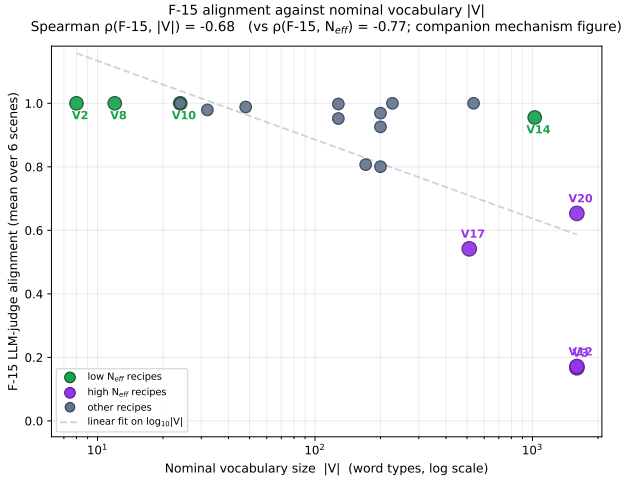
The V13–V20 recipes added in the P3 extension introduce three new interpretability profiles that complicate the V1–V12 story.

a) *V13 (VQ-VAE codebook) and V17 (sparse-coding dictionary):* Both recipes produce *opaque token alphabets*: the codeword index does not map to a wavelength, an absorption feature, or a band group. F-13 SHAP under V13 surfaces the same top codewords across topics (the codebook is shared across all classes during training), and a domain expert cannot translate those codewords into spectral semantics without a

TABLE III

WHAT THE TIE HANDLING IS WORTH (SIX-SCENE MEANS): THE REPORTED EXPECTATION OVER TIE ORDERS, THEN THE SINGLE-SORT RULE UNDER THREE ORDERS (THE ARCHIVED DESCENDING `NUMPY.ARGSORT`, BAND INDEX, AND RANDOM OVER FIVE SEEDS); THE EXPECTED OUTCOMES OVER THE 120 JUDGED DOCUMENTS (ALIGNED / MISALIGNED / AMBIGUOUS); THE NUMBER OF JUDGED DOCUMENTS WHOSE OUTCOME DEPENDS ON THE TIE ORDER; THE NUMBER WHOSE NON-ZERO COUNTS ARE ALL EQUAL; AND THE MEAN DOCUMENT LENGTH ON INDIAN PINES. V15 IS COMPUTED ON THE REGENERATED $Q = 8$ CORPUS.

Recipe	expectation	archived	index	random	E[outcomes]	tie-dep.	equal-count	tokens/doc
V2	1.000	1.000	1.000	1.000	120 / 0 / 0	0	0	200
V8	1.000	1.000	1.000	1.000	120 / 0 / 0	0	9	14
V9	1.000	1.000	1.000	1.000	84 / 0 / 36	0	120	1
V10	1.000	1.000	1.000	1.000	120 / 0 / 0	0	120	3
V15	0.988	0.992	—	—	118.6 / 1.4 / 0	4	120	6
V14	0.955	0.950	0.958	0.955	114.7 / 5.3 / 0	65	120	16
V13	0.952	0.950	0.967	0.957	114.3 / 5.8 / 0	7	120	4
V1	0.800	0.792	0.733	0.817	96.1 / 23.9 / 0	52	0	450
V20	0.653	0.642	0.467	0.667	78.4 / 41.6 / 0	97	0	1187
V12	0.171	0.158	0.067	0.173	20.6 / 99.4 / 0	120	120	200
V3	0.167	0.117	0.033	0.180	20.1 / 99.9 / 0	119	120	200



Same F-15 values and recipes as p5-dispersion-scatter: x-axis is nominal $|V|$ (from F15 [SON] instead of N_{eff}). V14 ($|V| = 1024$, 16 tokens per document) scores 0.95; V2, V8 and V9 sit at 1.0 by construction of the rule. V3 and V12 depend on the tie order of their equal-count documents (0.03 to 0.18).

Fig. 2. F-15 versus nominal vocabulary $|V|$ (log scale), nineteen recipes (Table II). A pure vocabulary-size account would place all points on a monotone decreasing curve. V14 ($|V| = 1024$, 16 tokens per document, F-15 = 0.96) and V20 ($|V| \approx 1376$, F-15 = 0.65) sit far above V3 / V12, so vocabulary size alone does not set F-15; V9 ($|V| = 647$, F-15 = 1.0) is fixed by the rule and is no evidence either way.

separate decoder visualisation. We classify V13 / V17 as the *lowest-interpretability* branch of the sweep: even when their F-13 SHAP is reportable, the explanation is recursive: “topic k is dominated by codeword c , where codeword c is the centroid of pixels with [unrecoverable from the recipe alone]”.

b) V14 (CWT-Morlet) and V18 (graph-Laplacian): The opposite case. V14 tokens are (`log_scale_idx`, `position_bucket`, `magnitude_bin`); each (scale, position) pair maps to a spectral region (one of eight position buckets) at a bandwidth set by the scale (sixteen scales from 1 to $B/2$ bands), and the bin quantises the wavelet magnitude. On Indian Pines and Pavia U the top SHAP tokens of every topic sit at the three coarsest scales (s_{13} to s_{15}), so V14 explanations name broad spectral shape in a region rather than narrow absorption features. V18 tokens are (`eigenvector_id`, `bin`); the eigenvector index names a mode of the kNN graph and the bin a position along it. The

top SHAP tokens of a V18 topic mix low- and high-order modes (on Pavia U, modes 0 to 14), so a V18 explanation names positions on graph modes, which carry no spectral meaning without a map of each mode back to pixels. V14 thus extends the V1 + V7 physically-grounded profile to multi-scale spectral shape; V18 is grounded in the pixel graph rather than in the spectrum.

c) V19 (UMAP coords) and V15 (spectral indices): V15 is the easiest: top tokens of every topic map directly to NDVI / MNDWI / NBR / NDSI / EVI / SAVI bins, with published semantics. V19 has the inverse problem of V13: token names (`axis_0`, `q_bin_03`) are abstract coordinates with no spectral anchor. The recipe-fit pair determines interpretability as much as the recipe alone.

d) V20 (MI-weighted bands): V20 inherits V3’s interpretability (its (band, q-bin) tokens are spectrally grounded) and adds a label-aware twist (10): high-MI bands emit more copies than low-MI bands (on Indian Pines 169 of the 200 bands receive at least 5 of the 8 possible copies and one band none), and its topics are less dispersed than V3’s and V12’s (N_{eff} 309 against 431 and 400, §VI). Its SHAP attributions are about as concentrated as V3’s: the top token carries 0.28 of the top-8 mean $|\text{SHAP}|$ mass, against 0.26 for V3 and 0.27 for V12. V20 keeps V3’s spectrally grounded vocabulary, adds label information and needs no codebook, which makes it the most readable of the extension recipes; its weights come from the labels of the same pixels, so its explanations are label-informed by construction.

e) F-22 counterfactual under the extension: Table I already reports F-22 for all seven extension recipes across the six labelled scenes at $Q = 8$: V20, V12 and V3 form the ultra-robust band (sentinel-patched means 26.3 / 24.5 / 23.5), though, as noted in Section V, that ordering is dominated by never-flipped scenes rather than a strict robustness gap. A full F-22 across the $Q = 16/32$ cross-product is left to a follow-up.

VIII. RECOMMENDATION

For interpretability reporting, we recommend the following ordering:

- 1) **F-13 SHAP attribution.** Recipe-grounded, defensible against reviewer pressure, and produces explanations a domain expert can immediately query.
- 2) **F-22 counterfactual robustness.** Complements F-2/F-7 by ruling out trivial-overfit explanations.
- 3) **F-15 LLM-judge.** Only with its construction effects reported beside it (§VI). The rule rewards small vocabularies and short documents and cannot judge one-token documents misaligned; its tie dependence is removed by taking the expectation over tie orders.

The V13–V20 extension (§VII) further suggests a recipe-as-interpretability-tier classification:

- **Tier I (physically-grounded):** V1, V7, V14, V20. Top SHAP tokens map to a wavelength, absorption feature, scale band, or label-weighted band. Use for any publishable interpretability claim.
- **Tier II (semantically-grounded):** V8, V10, V15. Top tokens map to endmember labels, group labels, or vegetation indices.
- **Tier III (learned-codebook-grounded):** V11, V12, V13, V17. Top tokens are codewords / GMM components / dictionary atoms. Interpretable only via a separate codebook-decoder visualisation.
- **Tier IV (manifold-coordinate):** V18, V19. Topics point to kNN-graph modes or UMAP coordinates with no spectral anchor. Use only as a sanity baseline.

IX. DISCUSSION

The three axes are complementary, not redundant. F-13, F-22 and F-15 probe different facets of the same LDA fit. F-13 answers *which tokens explain a topic assignment*; F-22 answers *how robust that assignment is*; F-15 answers *whether a human-like judge would agree the document belongs to its topic*, a question the current rule answers only through its construction effects (§VI). V20 shows how the axes can diverge: its topics are less dispersed (N_{eff} 309 against 431 for V3), its topics sit in the ultra-robust F-22 band (sentinel-patched mean = 26.3, with documents about six times longer than V3’s), and P3 finds it leads F-7 at $Q = 32$, while its SHAP attributions are no more concentrated than V3’s. Whether these reflect one property of the MI weighting is not tested here.

F-15 measures the metric, not the model. The central methodological lesson is that an automated alignment proxy can fail for reasons that have nothing to do with topic quality. The anti-correlation between F-15 and F-2 on V3 / V12 is not evidence that those recipes produce incoherent topics (P3 ranks V12 *first* on F-2 and F-7) but evidence that a top- N identity-overlap rule is governed by its construction: vocabulary size and tokens per document (§VI), and, unless ties are averaged out as they are here, the order of tied counts. An actual LLM oracle, given the same token lists and asked for identity overlap, would inherit the same effects; only a judge equipped with *semantic* comparison of token meanings (e.g. the spectral region a token names) would escape it. This is a cautionary result for the growing practice of using LLMs to score topic models [5]: the prompt must expose token *semantics*, not token

strings, or the score will track vocabulary engineering rather than topic content.

Interpretability is a recipe property, not an LDA property. The tier classification of §VIII shows that the same LDA backbone yields explanations ranging from directly physical (V1, V7, V14, V20) to recursively opaque or abstract (V13, V17, V18, V19). “LDA is interpretable” is therefore an incomplete claim; the defensible claim is “LDA on a physically-grounded wordification is interpretable.”

X. LIMITATIONS

LDA only. Every result here is for the LDA backbone (1). The closed-form posterior that makes F-13 tractable does not transfer unchanged to the two amortised neural backbones of P4: the model labelled ProdLDA, which is the mixture-decoder form (NVLDA) of Srivastava and Sutton’s amortised topic model [21] with a standard normal prior (3) (the product-of-experts decoder that defines ProdLDA was not run), and ETM [22] (4), which shares that prior and encoder and differs only in its embedding decoder; replicating these axes under P4’s backbones is open work.

Deterministic stand-in for the LLM oracle. F-15 was produced by a deterministic encoding of the alignment rule rather than by live per-call LLM judging. This makes the run reproducible and free, but it cannot exhibit the semantic-comparison behaviour that a true oracle might; the API builder (`build_v_sweep_f15_llm_alignment.py`) is provided for users who wish to substitute live judging. Our claims about F-15 concern the *metric definition* (17) and its inputs (the top- N token lists) and hold for any judge that receives those lists.

N_{eff} **is computed on ϕ_k .** We measure dispersion on the topic-word distributions, the quantity the topic side of the F-15 rule reads; the canonical metric (18) is stated on a generic mass vector p_d and applies equally to per-document counts, which the document side of the rule reads and which this study does not summarise.

Partial F-22 under the extension. The full F-22 table (Table I) is complete for the $Q = 8$ sweep, but the $\times Q \times$ extension-recipe cross-product (§VII) is profiled only spot-wise; a full re-run is left to a follow-up.

Six labelled scenes. The HIDSAG mineralogical subsets used in P1 [23] are not included here because their per-pixel label structure differs; the F-15 results should be re-checked on them before being claimed as scene-independent.

XI. CONCLUSION

We instantiated three post-hoc interpretability axes (F-13 SHAP attribution, F-22 counterfactual L_1 , and F-15 LLM-judge alignment) on the LDA wordification sweep of P3, and reported three results. First, F-13 SHAP gives recipe-grounded explanations whose physical meaning is fixed by the recipe and cannot silently shift across scenes; this is the axis we recommend as the primary interpretability artefact. Second, F-22 separates the recipes into an ultra-robust band (V20 / V12 / V3, sentinel-patched means 26.3 / 24.5 / 23.5, in which most documents never flip within 50 steps) and a fragile

regime ($V9 = 1.0$); F-22 grows with the document length, which this ordering partly reflects. Third, and most important methodologically, F-15 as defined is governed by construction effects of its top-10 overlap rule rather than by agreement between documents and topics: it is 1.0 by construction for small vocabularies ($V2$, $V8$) and one-token documents ($V9$), and near 1.0 for three- and four-token documents. For recipes whose documents hold every token once ($V3$, $V12$) no sort order is informative, so we report the expectation over tie orders (0.167 and 0.171) instead of one draw between 0.03 and 0.18. $V20$ stays far above $V3$ and $V12$ under every tie order and has less dispersed topics (N_{eff} 309 against 431 and 400), but its copies also rank the tokens of its documents, so the gap cannot be attributed to topic dispersion. We therefore recommend reporting F-15 only with these construction effects beside it, and we caution that LLM-as-judge topic evaluation inherits them unless the judge is given token *semantics* rather than token strings. Extending these axes across the P4 backbones, and re-checking the F-15 results on the HIDSAG subsets, are the natural next steps.

Declaration of generative AI use. Generative AI tools were used for formulating the alignment rule that stands in for the LLM oracle of the F-15 axis, applied as a deterministic function. The author reviewed and verified all content and takes full responsibility for the work.

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